

Relationships among FuncCPA and Its Related Notions [★]

Takumi Shinozaki¹[0009–0004–2377–3927],
Tatsuaki Okamoto^{1,2}[0000–0001–7803–812X],
Keisuke Tanaka¹[0000–0003–1330–4495],
Masayuki Tezuka¹[0009–0003–5260–3372], and
Yusuke Yoshida¹[0000–0003–0905–5641]

¹ Institute of Science Tokyo

shinozaki.t.9c3e@m.isct.ac.jp, keisuke@comp.isct.ac.jp,
tezuka.m.eab3@m.isct.ac.jp, yoshida.yusuke@comp.isct.ac.jp

² NTT Social Informatics Laboratories
tatsuaki.okamoto@gmail.com

Abstract. Akavia, Gentry, Halevi, and Vald (TCC’22, JoC’25) introduced the security notion of function-chosen-plaintext-attack (FuncCPA security) for public-key encryption schemes. FuncCPA is defined by adding a functional re-encryption oracle to the IND-CPA game. This notion is crucial for secure computation applications where the server is allowed to delegate a part of the computation to the client. Dodis, Halevi, and Wichs (TCC’23) introduced a stronger variant called FuncCPA⁺, and conjectured that FuncCPA⁺ is strictly stronger than FuncCPA, while they showed FuncCPA⁺ implies FuncCPA. Seeking insights into this conjecture, they showed that ReEncCPA⁺ is strictly stronger than ReEncCPA, where ReEncCPA and ReEncCPA⁺ are restricted versions of FuncCPA and FuncCPA⁺ respectively.

In this paper, contrary to their conjecture, we show that FuncCPA⁺ is equivalent to FuncCPA. We also introduce new variants of FuncCPA; WeakFuncCPA, OW-FuncCPA, and OW-WeakFuncCPA. WeakFuncCPA is a restricted variant of FuncCPA in that an oracle query is prohibited after the challenge query (like IND-CCA1). OW-FuncCPA and OW-WeakFuncCPA are the one-way (OW) versions of FuncCPA and WeakFuncCPA, respectively. This paper shows that WeakFuncCPA and OW-FuncCPA are equivalent to FuncCPA, that is, all of FuncCPA, FuncCPA⁺, WeakFuncCPA, and OW-FuncCPA are equivalent. Considering the separation of IND-CCA1 and IND-CCA2, and that of OW-CPA and IND-CPA, these results are surprising. To show the equivalence, we develop novel techniques to utilize functional re-encryption oracles. We then provide the separation results that OW-WeakFuncCPA does not imply FuncCPA and ReEncCPA⁺ does not imply FuncCPA.

[★] Part of this work was supported by JST K Program JPMJKP24U2, and JST CREST JPMJCR2113.

1 Introduction

1.1 Background

A homomorphic encryption scheme enables operations on encrypted data. Homomorphic encryption schemes achieve IND-CPA security, but cannot achieve IND-CCA2 security. However, IND-CPA security is not sufficient in certain application scenarios. Akavia, Gentry, Halevi, and Vald [1,2] focused on protocols in secure computation where a server delegates some of the processing back to a client holding the secret key. They showed that if the server is malicious, the client's secret information can be leaked even if the encryption scheme is IND-CPA secure. This vulnerability stems from the fact that IND-CPA security does not capture a situation where the malicious server requests some processing of a ciphertext to the client.

To address this issue, Akavia et al. proposed FuncCPA security for public-key encryption schemes. This security is defined by adding a *functional re-encryption oracle* to the IND-CPA security game. This oracle takes a function f and a ciphertext ct as input, decrypts ct , applies f to the result, and returns a new ciphertext $\text{Enc}(pk, f(\text{Dec}(sk, ct)))$. At first glance, this oracle seems useless to the adversary. However, they showed that a scheme exists that is IND-CPA secure but can be broken using this oracle.

Subsequently, Dodis, Halevi, and Wichs [4] proposed a variant of FuncCPA security called FuncCPA^+ security. This security is defined by a game where the adversary tries to distinguish between a functional re-encryption oracle and an oracle that always returns an encryption of plaintext 0. They showed that FuncCPA^+ security implies FuncCPA security, i.e. if an encryption scheme satisfies FuncCPA^+ security, then it also satisfies FuncCPA. They also showed that IND-CCA1 security implies FuncCPA^+ security. Furthermore, they proposed a construction of FuncCPA^+ secure scheme based on any IND-CPA secure scheme.

Dodis et al. introduced FuncCPA^+ security as a possibly stronger definition, but no separation has been found. While seeking insights into this problem, they found there exists an essential difference between them when the functions allowed to query the oracle are restricted to only the identity function. Namely, they proved ReEncCPA^+ security is strictly stronger than ReEncCPA security, where ReEncCPA^+ and ReEncCPA are the restricted versions of FuncCPA^+ and FuncCPA. Following this observation, they conjectured a separation between FuncCPA^+ and FuncCPA, and posed its proof as an open problem.

By focusing on the number of function inputs, Dodis et al. revealed a separation between FuncCPA security, which permits multi-input functions, and its more restricted single-input variants. To formalize this, they introduced 1- FuncCPA and 1- FuncCPA^+ security, where the oracle accepts only single-input functions. They found a scheme that is 1- FuncCPA^+ secure but not FuncCPA secure. This result highlights that the oracle's ability to process multiple inputs is a factor affecting security levels.

1.2 Some Previous Works on FuncCPA Security

Nuida [8] studied invalid-ciphertext queries in the context of FuncCPA security and proposed two variants, $\text{FuncCPA}^\dagger$ and $\text{FuncCPA}^{\dagger\dagger}$, that differ in how invalid ciphertexts are handled. Nuida observed that if the client simply rejects invalid queries, client privacy can be compromised even when the encryption scheme is FuncCPA secure. As a countermeasure, Nuida proposed “silent abort,” in which the client returns a dummy ciphertext (e.g., an encryption of 0) on invalid input. Nuida also showed that, under a fully homomorphic encryption (FHE) scheme that is IND-CPA secure, sanitizable, and circularly secure, this countermeasure ensures privacy against malicious servers.

Additionally, a multi-key variant of FuncCPA has been proposed. Nakajima, Hara, and Yamashita [7] introduced multi-key funcCPA (MKfuncCPA) security, extending FuncCPA to the multi-key setting. They showed that a KDM-secure public-key encryption scheme is MKfuncCPA-secure, and that analogous arguments apply to symmetric-key encryption.

In a related direction, Manulis and Nguyen [6] studied how fully homomorphic encryption (FHE) schemes can achieve security beyond IND-CCA1 and proposed vCCA. They proved that vCCA implies FuncCPA, and established a separation between FuncCPA and vCCA.

1.3 Motivation and Open Problems

Although the above results clarify several relationships between security notions related to FuncCPA security, there are many questions left unanswered. To understand the nature of FuncCPA security, it is important to explore the relationships among FuncCPA and its variants.

It would be natural to consider a variant of FuncCPA security that restricts access to the functional re-encryption oracle after receiving the challenge ciphertext. Similar variants are known in the context of CCA security. While the IND-CCA2 security game allows decryption queries after the challenge query, the IND-CCA1 security game prohibits such queries. As an analogy of IND-CCA1, we define WeakFuncCPA security as FuncCPA security game where access to the functional re-encryption oracle after the challenge query is prohibited. Since IND-CCA1 is a strictly weaker notion than IND-CCA2, it is expected a similar relationship holds between WeakFuncCPA security and FuncCPA security. Is WeakFuncCPA security *strictly* weaker than FuncCPA security?

We can also consider another variant of FuncCPA security focusing on the goal of the adversary. The two typical goals in security games of encryption schemes are indistinguishability (IND) and one-wayness (OW). While IND is considered in the context of FuncCPA security, OW is not. Thus we introduce OW-FuncCPA security by a security game where the adversary aims to break OW, given access to the functional re-encryption oracle. Since OW is strictly weaker security than IND under CPA or CCA setting, it is natural to ask whether OW-FuncCPA security is *strictly* weaker than FuncCPA security. A related question is whether OW-WeakFuncCPA security is *strictly* weaker than FuncCPA security. Here, OW-WeakFuncCPA is the OW version of WeakFuncCPA.

Another question is whether ReEncCPA^+ security is *strictly* weaker than 1- FuncCPA security. Prior to this work, no separation between ReEncCPA^+ and 1- FuncCPA was known.

In summary, we have the following questions about the relationships between these security notions related to FuncCPA .

- Is FuncCPA security *strictly* weaker than FuncCPA^+ security?
- Is WeakFuncCPA security *strictly* weaker than FuncCPA security?
- Is OW- FuncCPA security *strictly* weaker than FuncCPA security?
- Is OW- WeakFuncCPA security *strictly* weaker than FuncCPA security?
- Is ReEncCPA^+ security *strictly* weaker than 1- FuncCPA security?

1.4 Our Contributions

In this work, we explore and answer these questions. We enumerate our contributions as follows:

Theorem 1 WeakFuncCPA security implies FuncCPA^+ security .

Theorem 4 OW- FuncCPA security implies FuncCPA security.

Theorem 5 OW- WeakFuncCPA security does not imply FuncCPA security.

Theorem 6 ReEncCPA^+ security does not imply 1- FuncCPA security.

Theorems 1 and 4 answer the first three questions negatively (i.e. FuncCPA , FuncCPA^+ , WeakFuncCPA , and OW- FuncCPA are equivalent), while Theorems 5 and 6 answer the last two affirmatively.

We summarize the relationships among FuncCPA security and its related notions in Figure 1. For the proof of Theorem 1, we introduced an intermediate security notion $\Delta\text{FuncCPA}^+$. The left side of the figure illustrates the flow of the equivalence proofs. The right side of the figure shows the separations.

Our results significantly revise the current understanding of FuncCPA security. We resolve the Dodis et al. open problem by proving that FuncCPA and FuncCPA^+ security are equivalent, contrary to their conjecture. This result is a part of a comprehensive equivalence shown in Corollary 1, showing that FuncCPA , FuncCPA^+ , WeakFuncCPA , and OW- FuncCPA security are all equivalent. Unexpectedly, this result implies that neither the post-challenge restriction of WeakFuncCPA nor the one-way goal of OW- FuncCPA weakens FuncCPA security. This highlights the power of the functional re-encryption oracle.

Beyond these equivalences, we also found strictly weaker FuncCPA variants. Theorem 6, for instance, confirms that ReEncCPA , which restricts oracle functionality to only the identity function, is strictly weaker, highlighting the crucial role of versatile function application in FuncCPA . Furthermore, Theorem 5 shows that OW- WeakFuncCPA , a notion that combines post-challenge query restriction and a one-way goal, also results in a separation from FuncCPA security.

1.5 Technical Overview

This section outlines the core ideas behind our main technical contributions.

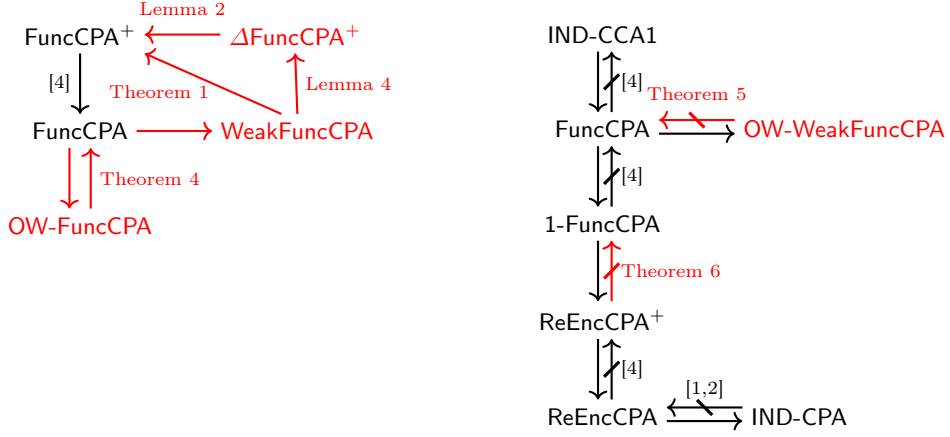


Fig. 1. FuncCPA and related security notions. A solid arrow $A \rightarrow B$ means any scheme satisfying A also satisfies B . A hatched arrow $B \nrightarrow A$ means there exists a scheme satisfying B but not A . Black arrows represent known results, red arrows our contributions.

WeakFuncCPA Security Implies FuncCPA⁺ Security Before explaining our main technique for Theorem 1, we explain a weaker result in Theorems 2 and 3, showing that FuncCPA implies FuncCPA⁺ when multiple-input functions, specifically those accepting at least two inputs, are allowed for the functional re-encryption oracle. Note that this result is also new and incomparable to the main result since the reduction is simple and efficient.

To prove this result, we must construct a reduction $\mathcal{B}$, which attacks the FuncCPA game, given an adversary $\mathcal{A}$ for the FuncCPA⁺ game. To accomplish the task of $\mathcal{B}$, (1) $\mathcal{B}$ sends $m_0 = 0$ and $m_1 = 1$ to the FuncCPA challenger $\mathcal{CH}$, and obtains the challenge ciphertext $ct^* = \text{Enc}(pk, b)$ from $\mathcal{CH}$, where b is $\mathcal{CH}$'s challenge bit. (2) $\mathcal{B}$ then tries to distinguish ct with b by using the power of $\mathcal{A}$. $\mathcal{B}$ plays the role of the FuncCPA⁺ challenger for $\mathcal{A}$. Let (ct, f) be a query from $\mathcal{A}$ to the challenger or $\mathcal{B}$. (3) $\mathcal{B}$ designs a two-input function g such that $g(x, 0) = 0$ and $g(x, 1) = x$. $\mathcal{B}$ sends $((ct, ct^*), g \circ (f, \text{id}))$ to $\mathcal{CH}$, where id is the identity function. (4) $\mathcal{CH}$ returns $\text{Enc}(pk, g(f(\text{Dec}(sk, ct)), b))$ to $\mathcal{B}$, which is $\text{Enc}(pk, 0)$ if $b = 0$, and $\text{Enc}(pk, f(\text{Dec}(sk, ct)))$ if $b = 1$. (5) $\mathcal{B}$ sends this value to $\mathcal{A}$, which is the correct FuncCPA⁺ oracle reply to $\mathcal{A}$'s query (ct^*, f^*) with using $\mathcal{CH}$'s challenge bit b . (6) Repeat this process with $\mathcal{A}$, and finally $\mathcal{A}$ outputs b' , which is $\mathcal{B}$'s output. Clearly, $\mathcal{B}$ attacks the FuncCPA game with the help of $\mathcal{A}$'s ability of attacking the FuncCPA⁺ game. See Section 3.3 for details of this reduction and its variants.

Unfortunately, this reduction technique does not apply to Theorem 1 for two reasons. First, the reduction essentially relies on the functional re-encryption oracle for multiple-input functions. In other words, the result only holds for $\ell \geq 2$ and it seems difficult to extend the proof to the case of $\ell = 1$. Second,

the reduction essentially use the functional re-encryption oracle after receiving the challenge ciphertext. Note that **WeakFuncCPA** security prohibits functional re-encryption queries after receiving the challenge ciphertext. Therefore, this technique does not give any insight into the relation to **WeakFuncCPA** security.

In the proof of our main theorem, we need a novel technique that works for single-input functions. When $\mathcal{B}$ wants to distinguish $ct^* = \text{Enc}(pk, m_b)$ with the help of $\mathcal{A}$ for the **FuncCPA**⁺, which queries (ct, f) , $\mathcal{B}$ should reply to $\mathcal{A}$ $\tilde{ct}_0 = \text{Enc}(pk, 0)$ if $b = 0$, and $\tilde{ct}_1 = \text{Enc}(pk, f(\text{Dec}(sk, ct)))$ if $b = 1$, where b is $\mathcal{CH}$'s challenge bit. If an oracle query function is single-input, $ct^* = \text{Enc}(pk, m_b)$ should be determined after ct is given. Otherwise, a two-input function for (ct, ct^*) would be necessary to produce $\tilde{ct}_b$. Then, a possible solution is that $ct^* = \text{Enc}(pk, m_b)$ is used for $\tilde{ct}_b$, when we set $m_1 = f(\text{Dec}(sk, ct))$ and $m_0 = 0$.

To realize this idea, $\mathcal{B}$ has to reveal $m_1 = f(\text{Dec}(sk, ct))$, after $\mathcal{A}$ queries (ct, f) . $\mathcal{B}$ designs a *single-input* function g_i such that $g_i(x) = 0$ if $[x]_i = 0$, and $g_i(x) = x$ if $[x]_i = 1$, where $[x]_i$ is the i -th bit of x . Then, $\mathcal{B}$ uses the functional re-encryption oracle with input $(ct, g_i \circ f)$, which outputs $\tilde{ct}_0 = \text{Enc}(pk, 0)$ if $[m_1]_i = 0$, and $\tilde{ct}_1 = \text{Enc}(pk, f(\text{Dec}(sk, ct)))$ if $[m_1]_i = 1$. $\mathcal{B}$ gives the oracle's output $\tilde{ct}_b$ to $\mathcal{A}$, which gives b' to $\mathcal{B}$. When $\mathcal{A}$ distinguishes $\tilde{ct}_b$ with a certain advantage, $\mathcal{B}$ repeats the process for i many times and obtains $[m_1]_i$ by the majority rule with overwhelming probability. Repeating the process for all bits of m_1 , $\mathcal{B}$ recovers m_1 .

$\mathcal{B}$ then sends (m_0, m_1) to the **FuncCPA** challenger $\mathcal{CH}$ with the challenge bit b , and obtains the challenge ciphertext $ct^* = \text{Enc}(pk, m_b)$. $\mathcal{B}$ distinguishes it with the help of $\mathcal{A}$, which distinguishes ct^* with a certain advantage. Note that $\mathcal{B}$ makes the functional re-encryption oracle queries only before the challenge ciphertext query. This is the reason why the reduction works for **WeakFuncCPA**.

The above explanation of the reduction is a key idea, and we need more details to complete the proof. For example, (1) we show how to utilize $\mathcal{A}$ many times for the same query (ct, f) . (2) Although $[m_1]_i$ is a fixed value given m_1 , the value should be uniformly distributed to analyze the majority rule by the Chernoff bound. We use a technique to mask it with a uniformly and independently random bit. (3) The previous discussion implicitly assumed that $\mathcal{A}$ makes the challenge query of (ct, f) only once, and distinguishes its ciphertext or a ciphertext of 0. However, **FuncCPA**⁺ adversary asks such queries an arbitrary number of times. To close this gap, we introduce $\Delta\text{FuncCPA}^+$ security, where the adversary makes the challenge query only once as in the above discussion. We show $\Delta\text{FuncCPA}^+$ security implies **FuncCPA**⁺ security via a hybrid argument. Then we use the above technique to prove **WeakFuncCPA** implies $\Delta\text{FuncCPA}^+$ security.

OW-FuncCPA Security Implies FuncCPA Security To prove this result in Theorem 4, we must construct a reduction $\mathcal{B}$, which attacks the **OW-FuncCPA** game, given an adversary $\mathcal{A}$ for the **FuncCPA** game. To accomplish the task of $\mathcal{B}$, (1) $\mathcal{B}$ plays the role of the **FuncCPA** challenger for $\mathcal{A}$. $\mathcal{B}$ replies $\mathcal{A}$'s queries to **FuncCPA**'s functional re-encryption oracle by using **OW-FuncCPA**'s oracle. (2) When $\mathcal{A}$ outputs challenge plaintexts (m_0, m_1) , $\mathcal{B}$ submits a challenge request

and receives a challenge ciphertext $ct^* = \text{Enc}(pk, m)$, m is randomly chosen from the message space. (3) $\mathcal{B}$ designs a function $g_i(x)$ such that $g_i(x) = m_0$ if $[x]_i = 0$, and $g_i(x) = m_1$ if $[x]_i = 1$. $\mathcal{B}$ queries (ct^*, g_i) to the oracle, and receives the reply $\tilde{ct}_i$, which is $\text{Enc}(pk, m_0)$ if $[m]_i = 0$, and $\text{Enc}(pk, m_1)$ if $[m]_i = 1$. (4) $\mathcal{B}$ gives $\tilde{ct}_i$ to $\mathcal{A}$ which distinguishes it with a certain advantage. (5) Repeat the process of (3) and (4) many times and obtain $[m]_i$ by the majority rule. (6) $\mathcal{B}$ recovers m by repeating this process for all i .

Note that this reduction essentially uses the functional re-encryption oracle after receiving the challenge ciphertext. Therefore, we only show that FuncCPA is equivalent to OW-FuncCPA, not OW-WeakFuncCPA. Indeed, we show there exists a separation between FuncCPA and OW-WeakFuncCPA.

ReEncCPA⁺ Security Does Not Imply 1-FuncCPA Security We show a separation between ReEncCPA⁺ and 1-FuncCPA in Theorem 6. For our separation, we modify an IND-CCA2 secure scheme $\mathcal{E}$ into a punctured scheme $\mathcal{E}'$ that is ReEncCPA⁺ secure, yet vulnerable to an attack with functional re-encryption. Our attack queries the functional re-encryption oracle using a simple non-identity function that sets the least significant bit to 1, and a special “trapdoor” ciphertext. When the query hits a punctured point in $\mathcal{E}'$, the oracle reveals its secret value. Despite this vulnerability, the encryption scheme $\mathcal{E}'$ remains ReEncCPA⁺ secure because this attack is infeasible as long as the oracle accepts only the identity function.

2 Preliminaries

In this section, we briefly specify standard terminology, notations, and definitions used throughout this paper.

Notations. For a positive integer n , $[n]$ represents the set $\{1, \dots, n\}$. For a finite set X , $x \xleftarrow{\$} X$ denotes sampling an element x uniformly at random from X . For an algorithm $\mathcal{A}$, $y \leftarrow \mathcal{A}(x)$ indicates that $\mathcal{A}$ outputs y given input x . λ denotes the security parameter. Probabilistic polynomial time is denoted by PPT. A function $f(\lambda)$ is negligible with respect to λ if it converges faster than $\frac{1}{\lambda^c}$ for all constants $c > 0$, and is denoted as $f(\lambda) = \text{negl}(\lambda)$. For a bit string $x \in \{0, 1\}^n$, let $[x]_i$ be the i -th bit of x . For $j \in [n]$ and bits $b_j \in \{0, 1\}$, let $\text{maj}(\{b_j\}_{j \in [n]}) \in \{0, 1\}$ be the majority function. In other words, $\text{maj}(\{b_j\}_{j \in [n]}) = 1$ if $|\{j \in [n] \mid b_j = 1\}| > n/2$ and 0 otherwise.

Lemma 1 (Chernoff Bound). *Let a random variable X follow the binomial distribution $B(n, p)$. For all $0 \leq \delta < 1$, the following holds:*

$$\Pr[X \leq (1 - \delta)pn] \leq \exp\left(-\frac{\delta^2}{2}pn\right).$$

2.1 Public-Key Encryption

In this section, we provide the basic definitions of public-key encryption and its associated security notions.

Definition 1. A public-key encryption scheme $\mathcal{E}$ over the message space $\mathcal{M}$ is a tuple of three PPT algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$ defined as follows:

- $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$: The key generation algorithm takes a security parameter 1^λ as input, and outputs a pair of public and secret keys (pk, sk) .
- $ct \leftarrow \text{Enc}(pk, m)$: The encryption algorithm takes a public key pk and a message $m \in \mathcal{M}$ as input, and outputs a ciphertext ct .
- $m' \leftarrow \text{Dec}(sk, ct)$: The decryption algorithm takes a secret key sk and a ciphertext ct as input, and outputs a message $m' \in \mathcal{M} \cup \{\perp\}$. In this paper, we assume that the decryption algorithm is deterministic.

Correctness: A public-key encryption scheme $\mathcal{E} = (\text{Gen}, \text{Enc}, \text{Dec})$ is correct if for any $m \in \mathcal{M}$,

$$\Pr[(pk, sk) \leftarrow \text{Gen}(1^\lambda), ct \leftarrow \text{Enc}(pk, m) : \text{Dec}(sk, ct) = m] = 1$$

holds.

Definition 2 (IND-CPA/IND-CCA1/IND-CCA2 Security). Let $\mathcal{E}$ be a public-key encryption scheme over the message space $\mathcal{M}$. We say that $\mathcal{E}$ is *atk* secure for a security notion $\text{atk} \in \{\text{IND-CPA}, \text{IND-CCA1}, \text{IND-CCA2}\}$ if for any PPT adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, $\Pr[\text{EXP}_{\mathcal{A}, \mathcal{E}}^{\text{atk}}(\lambda) = 1] \leq 1/2 + \text{negl}(\lambda)$, where the *atk* experiment $\text{EXP}_{\mathcal{A}, \mathcal{E}}^{\text{atk}}(\lambda)$ between a challenger $\mathcal{CH}$ and an adversary $\mathcal{A}$ is defined as follows:

1. $\mathcal{CH}$ runs $\text{Gen}(1^\lambda)$ to obtain a key-pair (pk, sk) and gives pk to $\mathcal{A}_1$.
2. $\mathcal{A}_1$ has access to an oracle $\mathcal{O}_1$.
3. $\mathcal{A}_1$ outputs a pair of messages $m_0, m_1 \in \mathcal{M}$ with $|m_0| = |m_1|$ and its state st .
4. $\mathcal{CH}$ chooses a random bit $b \in \{0, 1\}$, computes the challenge ciphertext $ct^* \leftarrow \text{Enc}(pk, m_b)$, and gives (ct^*, st) to $\mathcal{A}_2$.
5. $\mathcal{A}_2$ has access to an oracle $\mathcal{O}_2$.
6. $\mathcal{A}_2$ outputs a message $b' \in \{0, 1\}$. The experiment's output is 1 if $b = b'$ and 0 otherwise.

For IND-CPA security, the oracles $\mathcal{O}_1$ and $\mathcal{O}_2$ always return $\perp$. For IND-CCA1 security, $\mathcal{O}_1(ct)$ returns $\text{Dec}(sk, ct)$, and $\mathcal{O}_2$ always returns $\perp$. For IND-CCA2 security, $\mathcal{O}_1(ct)$ returns $\text{Dec}(sk, ct)$, and $\mathcal{O}_2$ is the same as $\mathcal{O}_1$, except it returns $\perp$ on the challenge ciphertext ct^* .

Definition 3 (OW-CCA1/OW-CCA2 Security). Let $\mathcal{E}$ be a public-key encryption scheme over the message space $\mathcal{M}$. We say that $\mathcal{E}$ is *atk* secure for a security notion $\text{atk} \in \{\text{OW-CCA1}, \text{OW-CCA2}\}$ if for any PPT adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, $\Pr[\text{EXP}_{\mathcal{A}, \mathcal{E}}^{\text{atk}}(\lambda) = 1] = \text{negl}(\lambda)$, where the *atk* experiment $\text{EXP}_{\mathcal{A}, \mathcal{E}}^{\text{atk}}(\lambda)$ between a challenger $\mathcal{CH}$ and an adversary $\mathcal{A}$ is defined as follows:

1. $\mathcal{CH}$ runs $\text{Gen}(1^\lambda)$ to obtain a key-pair (pk, sk) and gives pk to $\mathcal{A}_1$.
2. $\mathcal{A}_1$ has access to an oracle $\mathcal{O}_1$.
3. $\mathcal{A}_1$ outputs a challenge request and its state st .
4. $\mathcal{CH}$ chooses a message $m \leftarrow \mathcal{M}$ randomly, computes the challenge ciphertext $ct^* \leftarrow \text{Enc}(pk, m)$, and gives (ct^*, st) to $\mathcal{A}_2$.
5. $\mathcal{A}_2$ has access to an oracle $\mathcal{O}_2$.
6. $\mathcal{A}_2$ outputs a message $m' \in \mathcal{M}$. The experiment's output is 1 if $m = m'$ and 0 otherwise.

For OW-CCA1 security, the oracle $\mathcal{O}_1$ returns $\text{Dec}(sk, ct)$, and $\mathcal{O}_2$ always returns $\perp$. For OW-CCA2 security, $\mathcal{O}_1(ct)$ returns $\text{Dec}(sk, ct)$, and $\mathcal{O}_2$ is the same as $\mathcal{O}_1$, except it returns $\perp$ on the challenge ciphertext ct^* .

2.2 FuncCPA Security

Akavia, Gentry, Halevi, and Vald [1] introduced FuncCPA security as a notion for public-key encryption schemes. They defined this security by adding a functional re-encryption oracle to the IND-CPA security game.

Definition 4 (FuncCPA Security). Let $\mathcal{E}$ be a public-key encryption scheme with message space $\mathcal{M}$ and a family of functions $\mathcal{F} = \{f : (\mathcal{M} \cup \{\perp\})^\ell \rightarrow \mathcal{M} \mid \ell \in \mathbb{N}\}$. We say that a scheme $\mathcal{E}$ is **FuncCPA** secure with respect to $\mathcal{F}$ if for all PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ there exists a negligible function $\text{negl}(\cdot)$ such that:

$$\text{Adv}_{\mathcal{A}}^{\text{FuncCPA}}(\lambda) := 2 \cdot \left| \Pr[\text{EXP}_{\mathcal{A}, \mathcal{E}, \mathcal{F}}^{\text{FuncCPA}}(\lambda) = 1] - \frac{1}{2} \right| = \text{negl}(\lambda)$$

where the FuncCPA indistinguishability experiment $\text{EXP}_{\mathcal{A}, \mathcal{E}, \mathcal{F}}^{\text{FuncCPA}}(\lambda)$ between a challenger $\mathcal{CH}$ and an adversary $\mathcal{A}$ is defined as follows:

1. $\mathcal{CH}$ runs $\text{Gen}(1^\lambda)$ to obtain a key-pair (pk, sk) . $\mathcal{CH}$ gives pk to $\mathcal{A}_1$.
2. $\mathcal{A}_1$ is given access to the following functional re-encryption oracle:
$$\text{reEnc}(pk, sk, (ct_1, \dots, ct_\ell), f) = \text{Enc}(pk, f(\text{Dec}(sk, ct_1), \dots, \text{Dec}(sk, ct_\ell)))$$
3. $\mathcal{A}_1$ outputs a pair of messages m_0, m_1 with $|m_0| = |m_1|$ and a state st .
4. $\mathcal{CH}$ chooses a random bit $b \in \{0, 1\}$, computes the challenge ciphertext $ct^* \leftarrow \text{Enc}(pk, m_b)$, and gives (ct^*, st) to $\mathcal{A}_2$.
5. $\mathcal{A}_2$ retains access to the oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$.
6. $\mathcal{A}_2$ outputs a bit b' . The experiment's output is 1 if $b = b'$, and 0 otherwise.

Note that the state st contains all the information obtained by $\mathcal{A}_1$.

Within the framework of FuncCPA security, we consider several restrictions on the function family $\mathcal{F}$ or the game.

- **FuncCPA.** When we omit to specify the function family $\mathcal{F}$, it is understood to be the set of all efficiently computable functions.
- **1-FuncCPA.** We say that a scheme is 1-FuncCPA secure if it is FuncCPA secure with respect to the family of all 1-input functions.

- **ReEncCPA**. We say that a scheme is ReEncCPA secure if it is FuncCPA secure with respect to the identity function.
- **WeakFuncCPA**. We define the WeakFuncCPA game by modifying the FuncCPA security game: the functional re-encryption oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$ returns $\perp$ for any query made by $\mathcal{A}_2$.

2.3 FuncCPA⁺ Security

Dodis, Halevi, and Wichs [4] introduced FuncCPA⁺ security for public-key encryption schemes. They defined FuncCPA⁺ security by an experiment that distinguishes between an oracle providing functional re-encryption and an oracle always returning a ciphertext of 0.

Definition 5 (FuncCPA⁺ Security). Let $\mathcal{E}$ be a public-key encryption scheme with message space $\mathcal{M}$ and a family of functions $\mathcal{F} = \{f : (\mathcal{M} \cup \{\perp\})^\ell \rightarrow \mathcal{M} \mid \ell \in \mathbb{N}\}$. We say that a scheme $\mathcal{E}$ is FuncCPA⁺ secure with respect to $\mathcal{F}$ if for all PPT adversaries $\mathcal{A}$ there exists a negligible function $\text{negl}(\cdot)$ such that:

$$\text{Adv}_{\mathcal{A}}^{\text{FuncCPA}^+}(\lambda) := 2 \cdot \left| \Pr[\text{EXP}_{\mathcal{A}, \mathcal{E}, \mathcal{F}}^{\text{FuncCPA}^+}(\lambda) = 1] - \frac{1}{2} \right| = \text{negl}(\lambda)$$

where the FuncCPA⁺ experiment $\text{EXP}_{\mathcal{A}, \mathcal{E}, \mathcal{F}}^{\text{FuncCPA}^+}(\lambda)$ between a challenger $\mathcal{CH}$ and an adversary $\mathcal{A}$ is defined as follows:

1. $\mathcal{CH}$ runs $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$ and chooses a random bit b . $\mathcal{CH}$ gives pk to $\mathcal{A}$.
2. $\mathcal{A}$ is given access to the following functional re-encryption oracle:

$$\text{reEnc}_b(pk, sk, (ct_1, \dots, ct_\ell), f) = \begin{cases} \text{Enc}(pk, f(\text{Dec}(sk, ct_1), \dots, \text{Dec}(sk, ct_\ell))) & \text{if } b = 1 \\ \text{Enc}(pk, 0) & \text{if } b = 0 \end{cases}$$

3. $\mathcal{A}$ outputs a bit b' . The experiment's output is 1 if $b' = b$, and 0 otherwise.

Analogously, we define several restrictions for FuncCPA⁺ security.

- **FuncCPA⁺**. When we omit to specify the function family $\mathcal{F}$, it is understood to be the set of all efficiently computable functions.
- **1-FuncCPA⁺**. We say that a scheme is 1-FuncCPA⁺ secure if it is FuncCPA⁺ secure with respect to the family of all 1-input functions.
- **ReEncCPA⁺**. We say that a scheme is ReEncCPA⁺ secure with respect to the identity function.

3 FuncCPA Implies FuncCPA⁺

In this section, we prove WeakFuncCPA implies FuncCPA⁺ in two steps. First, we introduce an intermediate security notion, called $\Delta\text{FuncCPA}^+$. This notion lies between FuncCPA and FuncCPA⁺. $\Delta\text{FuncCPA}^+$ is defined so that it implies FuncCPA⁺ via a hybrid argument. It remains to show that WeakFuncCPA implies $\Delta\text{FuncCPA}^+$.

3.1 An Intermediate Definition Between FuncCPA and FuncCPA⁺

Definition 6 ($\Delta\text{FuncCPA}^+$ Security). Let $\mathcal{E}$ be a public-key encryption scheme with message space $\mathcal{M}$ and a family of functions $\mathcal{F} = \{f : (\mathcal{M} \cup \{\perp\})^\ell \rightarrow \mathcal{M} \mid \ell \in \mathbb{N}\}$. We say that $\mathcal{E}$ is $\Delta\text{FuncCPA}^+$ -secure with respect to $\mathcal{F}$ if for all PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ there exists a negligible function $\text{negl}(\cdot)$ such that

$$\text{Adv}_{\mathcal{A}}^{\Delta\text{FuncCPA}^+}(\lambda) := 2 \cdot \left| \Pr[\text{EXP}_{\mathcal{A}, \mathcal{E}, \mathcal{F}}^{\Delta\text{FuncCPA}^+}(\lambda) = 1] - \frac{1}{2} \right| = \text{negl}(\lambda),$$

where the experiment $\text{EXP}_{\mathcal{A}, \mathcal{E}, \mathcal{F}}^{\Delta\text{FuncCPA}^+}(\lambda)$ between a challenger $\mathcal{CH}$ and an adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ is defined as follows:

1. $\mathcal{CH}$ runs $\text{Gen}(1^\lambda)$ to obtain a key pair (pk, sk) and sends pk to $\mathcal{A}_1$.
2. $\mathcal{A}_1$ has oracle access to the functional re-encryption oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$.
3. $\mathcal{A}_1$ finally outputs a function $f \in \mathcal{F}$, ciphertexts $ct_1, \dots, ct_\ell$, where ℓ equals the number of inputs of f , and a state st .
4. $\mathcal{CH}$ chooses a random bit $b \in \{0, 1\}$ and sets

$$ct^* \leftarrow \begin{cases} \text{Enc}(pk, 0) & \text{if } b = 0, \\ \text{reEnc}(pk, sk, (ct_1, \dots, ct_\ell), f) & \text{if } b = 1, \end{cases}$$

and then sends (ct^*, st) to $\mathcal{A}_2$.

5. $\mathcal{A}_2$ outputs a bit b' . The experiment outputs 1 if $b' = b$, and 0 otherwise.

We show this variant implies the original FuncCPA⁺ security definition.

Lemma 2. $\Delta\text{FuncCPA}^+$ security implies FuncCPA⁺ security.

Proof. This follows from the standard hybrid argument. Let Q be the number of queries asked by the adversary in the FuncCPA⁺ game. For $i = 0, \dots, Q$, define the hybrid experiments H_i as follows:

H_i : This experiment is similar to $\text{EXP}_{\mathcal{A}, \mathcal{E}}^{\text{FuncCPA}^+}(\lambda)$, but the first i queries are answered by $\text{reEnc}(pk, sk, \cdot, \cdot)$, and the remaining queries are answered by $\text{Enc}(pk, 0)$.

Note that H_0 and H_Q are exactly the same as $\text{EXP}_{\mathcal{A}, \mathcal{E}}^{\text{FuncCPA}^+}(\lambda)$ in the case of $b = 0$ and $b = 1$ respectively.

Lemma 3. For $i \in [Q]$, there exists a reduction $\mathcal{B}_i$ such that the following holds:

$$|\Pr[1 \leftarrow \mathcal{A} \text{ in } H_{i-1}] - \Pr[1 \leftarrow \mathcal{A} \text{ in } H_i]| \leq \text{Adv}_{\mathcal{B}_i}^{\Delta\text{FuncCPA}^+}(\lambda).$$

Proof. The reduction algorithm $\mathcal{B}_i$ is defined as follows:

1. $\mathcal{B}_i$ receives pk and forwards it to the adversary $\mathcal{A}$.
2. For each oracle query $(ct_1, \dots, ct_\ell, f)$ from $\mathcal{A}$, let $j \in [Q]$ be the index of this query.

- (a) If $j < i$, $\mathcal{B}_i$ forwards the query to the challenger $\mathcal{CH}$ and relays the response ct' to $\mathcal{A}$
 - (b) If $j = i$, $\mathcal{B}_i$ treats it as the challenge query, forwards it to $\mathcal{CH}$, and replays the challenge ciphertext ct^* to $\mathcal{A}$.
 - (c) If $j > i$, $\mathcal{B}_i$ answers the query by returning $\text{Enc}(pk, 0)$
3. Finally, $\mathcal{B}_i$ outputs the output of $\mathcal{A}$.

If $b = 1$ in $\text{EXP}_{\mathcal{B}_i, \mathcal{E}}^{\Delta\text{FuncCPA}^+}$, the reduction perfectly simulates H_{i-1} . If $b = 0$, it perfectly simulates H_i . Therefore, the lemma holds. $\square$

Then by the triangle inequality and Lemma 3 we have

$$\begin{aligned}
\text{Adv}_{\mathcal{A}}^{\text{FuncCPA}^+}(\lambda) &= |\Pr[1 \leftarrow \mathcal{A} \text{ in } H_0] - \Pr[1 \leftarrow \mathcal{A} \text{ in } H_Q]| \\
&\leq \sum_{i \in [Q]} |\Pr[1 \leftarrow \mathcal{A} \text{ in } H_{i-1}] - \Pr[1 \leftarrow \mathcal{A} \text{ in } H_i]| \\
&\leq \sum_{i \in [Q]} \text{Adv}_{\mathcal{B}_i}^{\Delta\text{FuncCPA}^+}(\lambda) \leq Q \cdot \text{Adv}_{\mathcal{B}_Q}^{\Delta\text{FuncCPA}^+}(\lambda),
\end{aligned}$$

which completes the proof of Lemma 2. $\square$

3.2 Main Theorem

We are now ready to prove our main theorem.

Theorem 1. *WeakFuncCPA security implies FuncCPA⁺ security.*

The theorem follows from the following two lemmas, Lemma 4 and Lemma 2.

Lemma 4. *WeakFuncCPA security implies $\Delta\text{FuncCPA}^+$ security.*

Proof. For simplicity, we present the proof for $\ell = 1$. The case $\ell \geq 2$ follows by the same argument.

Reduction algorithm Let $\mathcal{M}$ be the message space and its message length is represented by a polynomial μ . Using a $\Delta\text{FuncCPA}^+$ adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, we construct a reduction algorithm $\mathcal{B}$ against WeakFuncCPA as follows:

1. $\mathcal{B}$ receives pk and forwards it to $\mathcal{A}_1$.
2. For each oracle query (ct_i, f_i) from $\mathcal{A}_1$, $\mathcal{B}$ forwards it to the WeakFuncCPA challenger $\mathcal{CH}$ and relays to $\mathcal{A}_1$ the reply $\text{Enc}(pk, f_i(\text{Dec}(sk, ct_i)))$, using fresh independent randomness in Enc each time.
3. $\mathcal{B}$ receives (ct, f, st) from $\mathcal{A}_1$.
4. For each $i \in [\mu]$, $\mathcal{B}$ recovers the i -th bit of $f(\text{Dec}(sk, ct))$ as follows. Let n be a repetition parameter that is polynomial in λ ; its value is fixed later in the analysis.

- (a) For each $j \in [n]$, sample an independent mask $\beta_{ij} \leftarrow \{0, 1\}$ and define

$$g_{ij}(x) = \begin{cases} 0 & \text{if } [x]_i \oplus \beta_{ij} = 0, \\ x & \text{if } [x]_i \oplus \beta_{ij} = 1. \end{cases}$$

- (b) Using the **WeakFuncCPA** challenger's oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$, compute

$$ct_{ij}^* \leftarrow \text{reEnc}(pk, sk, ct, g_{ij} \circ f),$$

where each call uses fresh independent randomness r_{ij} .

- (c) Run the adversary $b'_{ij} \leftarrow \mathcal{A}_2(ct_{ij}^*, st)$ and set the unmasked bit $b_{ij} \leftarrow b'_{ij} \oplus \beta_{ij}$.
- (d) Let the guess for the i -th bit be $[m_1]_i \leftarrow \text{maj}(\{b_{ij}\}_{j \in [n]})$.
5. Set $m_0 \leftarrow 0^\mu$ and $m_1 \leftarrow [m_1]_1 \parallel \dots \parallel [m_1]_\mu$. Then $\mathcal{B}$ sends (m_0, m_1) to $\mathcal{CH}$ and receives the challenge ciphertext ct^* .
6. $\mathcal{B}$ runs $\mathcal{A}_2$ with the *double-run trick* [3,5] and outputs $\tilde{b} \leftarrow \tilde{\mathcal{A}}_2(ct^*, st)$, where $\tilde{\mathcal{A}}_2$ is defined as follows:
- First run on the real challenge: $b' \leftarrow \mathcal{A}_2(ct^*, st)$.
 - Sample another challenge bit $\hat{b} \leftarrow \{0, 1\}$ and compute $\hat{ct}^* \leftarrow \text{Enc}(pk, m_{\hat{b}})$ using fresh independent randomness.
 - Second run on the simulated challenge: $\hat{b}' \leftarrow \mathcal{A}_2(\hat{ct}^*, st)$.
 - Output $\tilde{b} \leftarrow b' \oplus \hat{b} \oplus \hat{b}'$.

Analysis of the success probability Let $\varepsilon := \text{Adv}_{\mathcal{A}}^{\Delta\text{FuncCPA}^+}(\lambda)$ be the advantage of $\mathcal{A}$. Without loss of generality, we assume³

$$\Pr_{r,b,R}[b = \mathcal{A}_2(\text{Enc}(pk, m_b; r), st)] = \frac{1}{2} + \frac{\varepsilon}{2}.$$

Here, r denotes the randomness used to encrypt the challenge ciphertext. Let R denote the collection of all remaining randomness used throughout the $\Delta\text{FuncCPA}^+$ game: the randomness used by Gen ; the randomness used by all functional re-encryption oracles $\text{reEnc}(pk, sk, \cdot, \cdot)$ except for the challenge query; the randomness used by $\mathcal{A}_1$; and the randomness used by $\mathcal{A}_2$. Note that st includes the randomness of $\mathcal{A}_2$. We define the subset of $\{R\}$ as follows:

$$\text{Good} = \left\{ R \mid \Pr_{r,b}[b = \mathcal{A}_2(\text{Enc}(pk, m_b; r), st)] \geq \frac{1}{2} + \frac{\varepsilon}{4} \right\},$$

where $m_0 = 0^\mu$, $m_1 = f(\text{Dec}(sk, ct))$.

Lemma 5.

$$\Pr_R[R \in \text{Good}] \geq \frac{\varepsilon}{4}.$$

³ Although $\Pr_{r,b,R}[b = \mathcal{A}_2(\text{Enc}(pk, m_b; r), st)]$ could be $\frac{1}{2} - \frac{\varepsilon}{2}$ in general; we assume it is $\frac{1}{2} + \frac{\varepsilon}{2}$. To be precise, we can correct the sign of $\frac{\varepsilon}{2}$ and ensure it is $\frac{1}{2} + \frac{\varepsilon}{2}$ by providing it as advice to the non-uniform algorithm. Alternatively, without the advice, we can correct the sign by a random guess, which succeeds with probability $1/2$.

Proof.

$$\begin{aligned}
\frac{1}{2} + \frac{\varepsilon}{2} &= \Pr_{r,b}[b = \mathcal{A}_2(\text{Enc}(pk, m_b; r), st)] \\
&= \Pr_R[R \in \text{Good}] \Pr_{r,b}[b = \mathcal{A}_2(\text{Enc}(pk, m_b; r), st) \mid R \in \text{Good}] \\
&\quad + \Pr_R[R \notin \text{Good}] \Pr_{r,b}[b = \mathcal{A}_2(\text{Enc}(pk, m_b; r), st) \mid R \notin \text{Good}] \\
&\leq \Pr_R[R \in \text{Good}] \cdot 1 + \left(1 - \Pr_R[R \in \text{Good}]\right) \cdot \left(\frac{1}{2} + \frac{\varepsilon}{4}\right)
\end{aligned}$$

Subtracting $\frac{1}{2} + \frac{\varepsilon}{4}$ from both sides completes the proof. $\square$

Next, we show if the randomness is good, the reduction recovers $m_1 = f(\text{Dec}(sk, ct))$ correctly in step 4, except with negligible probability.

Lemma 6. *Let $n = \varepsilon^{-2}\omega(\log \lambda)$, the following holds:*

$$\Pr_{\{r_{ij}, \beta_{ij}\}_{i \in [\mu], j \in [n]}}[m_1 \neq f(\text{Dec}(sk, ct)) \mid R \in \text{Good}] \leq \mu \cdot \exp\left(-\frac{\varepsilon^2}{8(2+\varepsilon)}n\right).$$

Proof. The probability of failing to recover the entire $f(\text{Dec}(sk, ct))$ in step 4 is bounded by the union bound of the probabilities of failing to recover each $f(\text{Dec}(sk, ct))$ in step (d). Formally,

$$\begin{aligned}
&\Pr_{\{r_{ij}, \beta_{ij}\}_{i \in [\mu], j \in [n]}}[m_1 \neq f(\text{Dec}(sk, ct)) \mid R \in \text{Good}] \\
&\leq \sum_{i=1}^{\mu} \Pr_{\{r_{ij}, \beta_{ij}\}_{j \in [n]}}[\text{maj}(\{b_{ij}\}_{j \in [n]}) \neq [f(\text{Dec}(sk, ct))]_i \mid R \in \text{Good}]
\end{aligned}$$

holds. Since we condition on $R \in \text{Good}$, $\mathcal{A}_2$ guesses the random bit $[f(\text{Dec}(sk, ct))]_i \oplus \beta_{ij}$ and $[f(\text{Dec}(sk, ct))]_i$ with the following probability

$$\Pr_{r_{ij}, \beta_{ij}}[[f(\text{Dec}(sk, ct))]_i = b_{ij}] = \Pr_{r_{ij}, \beta_{ij}}[[f(\text{Dec}(sk, ct))]_i \oplus \beta_{ij} = b'_{ij}] \geq \frac{1}{2} + \frac{\varepsilon}{4}.$$

Therefore, the probability of failing to recover $f(\text{Dec}(sk, ct))_i$ is bounded by the Chernoff bound as follows:

$$\begin{aligned}
&\Pr_{\{r_{ij}, \beta_{ij}\}_{j \in [n]}}[\text{maj}(\{b_{ij}\}_{j \in [n]}) \neq [f(\text{Dec}(sk, ct))]_i] \\
&= \Pr_{\{r_{ij}, \beta_{ij}\}_{j \in [n]}}[|\{j \in [n] \mid b_{ij} = f(\text{Dec}(sk, ct))\}| \leq \frac{n}{2}] \\
&\leq \Pr_{\{r_{ij}, \beta_{ij}\}_{j \in [n]}}\left[B\left(n, \frac{1}{2} + \frac{\varepsilon}{4}\right) \leq (1 - \delta) \left(\frac{1}{2} + \frac{\varepsilon}{4}\right)n\right], \left(\delta = \frac{\varepsilon}{2 + \varepsilon}\right) \\
&\leq \exp\left(-\frac{\delta^2}{2} \left(\frac{1}{2} + \frac{\varepsilon}{4}\right)n\right) = \exp\left(-\frac{\varepsilon^2}{8(2 + \varepsilon)}n\right).
\end{aligned}$$

Therefore, the probability of the reduction algorithm $\mathcal{B}$ fails to recover m_1 is bounded by

$$\Pr_{\{r_{ij}, \beta_{ij}\}_{i \in [\mu], j \in [n]}} [m_1 \neq f(\text{Dec}(sk, ct)) \mid R \in \text{Good}] \leq \mu \cdot \exp\left(-\frac{\varepsilon^2}{8(2 + \varepsilon)}n\right).$$

□

Next, we bound the success probability of the reduction when a good state is chosen and the recovery of $f(\text{Dec}(sk, ct))$ succeeds.

Lemma 7. $\Pr[b = \tilde{b} \mid R \in \text{Good} \wedge m_1 = f(\text{Dec}(sk, ct))] \geq \frac{1}{2} + \varepsilon^2/8$

Proof. Let us denote

$$\Pr_{r,b}[b = A_2(\text{Enc}(pk, m_b; r), st) \mid R \in \text{Good} \wedge m_1 = f(\text{Dec}(sk, ct))] = \frac{1}{2} + \varepsilon_{\text{Good}}.$$

Note that $\varepsilon_{\text{Good}} \geq \varepsilon/4$ holds by the definition of **Good**. Then, under the condition $R \in \text{Good} \wedge m_1 = f(\text{Dec}(sk, ct))$, we have

$$\begin{aligned} & \Pr[b = b' \oplus \hat{b} \oplus \hat{b}'] \\ &= \Pr[b = b'] \cdot \Pr[\hat{b} = \hat{b}'] + \Pr[b \neq b'] \cdot \Pr[\hat{b} \neq \hat{b}'] \\ &= \left(\frac{1}{2} + \varepsilon_{\text{Good}}\right) \cdot \left(\frac{1}{2} + \varepsilon_{\text{Good}}\right) + \left(\frac{1}{2} - \varepsilon_{\text{Good}}\right) \cdot \left(\frac{1}{2} - \varepsilon_{\text{Good}}\right) \\ &= \frac{1}{2} + 2\varepsilon_{\text{Good}}^2 \geq \frac{1}{2} + \varepsilon^2/8. \end{aligned}$$

□

We also bound the success probability when the state is not good or the recovery of $f(\text{Dec}(sk, ct))$ failed.

Lemma 8. $\Pr[b = \tilde{b} \mid R \notin \text{Good} \vee m_1 \neq f(\text{Dec}(sk, ct))] \geq \frac{1}{2}$

Proof. In this case, if we denote

$$\Pr_{r,b}[b = A_2(\text{Enc}(pk, m_b; r), st) \mid R \notin \text{Good} \vee m_1 \neq f(\text{Dec}(sk, ct))] = \frac{1}{2} + \varepsilon_{\text{Bad}},$$

ε_{Bad} could be negative. However, by a similar argument as in the previous lemma, we have

$$\Pr[b = b' \oplus \hat{b} \oplus \hat{b}' \mid R \notin \text{Good} \vee m_1 \neq f(\text{Dec}(sk, ct))] \geq \frac{1}{2} + 2\varepsilon_{\text{Bad}}^2 \geq \frac{1}{2}.$$

□

Now we are ready to bound the success probability of $\mathcal{B}$ as follows:

$$\begin{aligned}
& \Pr[\text{EXP}_{\mathcal{B}, \mathcal{E}}^{\text{WeakFuncCPA}}(\lambda) = 1] \\
&= \Pr[R \in \text{Good} \wedge m_1 = f(\text{Dec}(sk, ct))] \Pr[b = \tilde{b} \mid R \in \text{Good} \wedge m_1 = f(\text{Dec}(sk, ct))] \\
&\quad + \Pr[R \notin \text{Good} \vee m_1 \neq f(\text{Dec}(sk, ct))] \Pr[b = \tilde{b} \mid R \notin \text{Good} \vee m_1 \neq f(\text{Dec}(sk, ct))] \\
&\geq \Pr[R \in \text{Good} \wedge m_1 = f(\text{Dec}(sk, ct))] \cdot \left(\frac{1}{2} + \frac{\varepsilon^2}{8} \right) \\
&\quad + \Pr[R \notin \text{Good} \vee m_1 \neq f(\text{Dec}(sk, ct))] \cdot \frac{1}{2} \\
&= \frac{1}{2} + \Pr[R \in \text{Good}] \Pr[m_1 = f(\text{Dec}(sk, ct)) \mid R \in \text{Good}] \cdot \frac{\varepsilon^2}{8} \\
&\geq \frac{1}{2} + \frac{\varepsilon^3}{32} \left(1 - \mu \cdot \exp \left(-\frac{\varepsilon^2}{8(2+\varepsilon)} n \right) \right).
\end{aligned}$$

Therefore, it follows that

$$\text{Adv}_{\mathcal{A}}^{\Delta \text{FuncCPA}^+}(\lambda) \leq \sqrt[3]{16 \cdot \text{Adv}_{\mathcal{B}}^{\text{WeakFuncCPA}}(\lambda) + \text{negl}(\lambda)}.$$

This completes the proof of Lemma 4. $\square$

Combining Lemma 4 and Lemma 2, we conclude that WeakFuncCPA secure schemes are also FuncCPA^+ secure.

3.3 Alternative Reductions

In this section, we give two alternative reductions showing that FuncCPA security implies FuncCPA^+ security. Theorem 2 shows that for $\ell \geq 2$, each adversarial query can be simulated using two oracle queries.

Theorem 2. *For any $\ell \geq 2$, ℓ -FuncCPA security implies ℓ -FuncCPA $^+$ security.*

Proof. Using a ℓ -FuncCPA $^+$ adversary $\mathcal{A}$, we construct a reduction algorithm $\mathcal{B}$ against ℓ -FuncCPA as follows:

1. $\mathcal{B}$ receives pk .
2. $\mathcal{B}$ sends $m_0 = 0, m_1 = 1$ to the ℓ -FuncCPA challenger $\mathcal{CH}$ and receives the challenge ciphertext ct^* .
3. $\mathcal{B}$ runs $\mathcal{A}$ with pk .
4. For each oracle query $(ct_1, \dots, ct_\ell, f)$ from $\mathcal{A}$, $\mathcal{B}$ answers as follows:
 - (a) Forwards the query to $\mathcal{CH}$ and obtains $ct' \leftarrow \text{reEnc}(pk, sk, (ct_1, \dots, ct_\ell), f)$.
 - (b) Defines

$$g(x_1, x_2) = \begin{cases} x_1 & \text{if } x_2 = 1, \\ 0 & \text{if } x_2 = 0 \end{cases}$$

and computes $ct'' \leftarrow \text{reEnc}(pk, sk, (ct', ct^*), g)$.

- (c) Answers $\mathcal{A}$ with ct'' .
- 5. $\mathcal{B}$ outputs $b' \leftarrow \mathcal{A}$.

The simulation is perfect because when $b = 0$ we obtain $ct'' = \text{Enc}(pk, 0)$, whereas when $b = 1$ we obtain $ct'' = \text{Enc}(pk, f(ct_1, \dots, ct_\ell))$. Consequently, the advantage of $\mathcal{B}$ in the ℓ -FuncCPA game equals that of $\mathcal{A}$ in the ℓ -FuncCPA⁺ game. $\square$

Theorem 3 shows that the query count remains fixed, while the number of inputs of the function submitted by the adversary increases from ℓ to $\ell + 1$.

Theorem 3. *For $\ell \geq 1$, $(\ell + 1)$ -FuncCPA security implies ℓ -FuncCPA⁺ security.*

Proof. Using a ℓ -FuncCPA⁺ adversary $\mathcal{A}$, we construct a reduction algorithm $\mathcal{B}$ against $(\ell + 1)$ -FuncCPA as follows:

1. $\mathcal{B}$ receives pk .
2. $\mathcal{B}$ sends $m_0 = 0, m_1 = 1$ to the $(\ell + 1)$ -FuncCPA challenger $\mathcal{CH}$ and receives the challenge ciphertext ct^* .
3. $\mathcal{B}$ runs $\mathcal{A}$ with pk .
4. For each oracle query $(ct_1, \dots, ct_\ell, f)$ from $\mathcal{A}$, $\mathcal{B}$ answers as follows:
 - (a) Defines

$$g(x_1, \dots, x_\ell, x_{\ell+1}) = \begin{cases} f(x_1, \dots, x_\ell) & \text{if } x_{\ell+1} = 1, \\ 0 & \text{if } x_{\ell+1} = 0 \end{cases}$$

and computes $ct' \leftarrow \text{reEnc}(pk, sk, (ct_1, \dots, ct_\ell, ct^*), g)$.

- (b) Answers $\mathcal{A}$ with ct' .
5. $\mathcal{B}$ outputs $b' \leftarrow \mathcal{A}$.

The simulation is perfect. If $b = 0$, then $ct' = \text{Enc}(pk, 0)$. If $b = 1$, then $ct' = \text{Enc}(pk, f(ct_1, \dots, ct_\ell))$. Therefore, the advantage of $\mathcal{B}$ in the $(\ell + 1)$ -FuncCPA game equals the advantage of $\mathcal{A}$ in the ℓ -FuncCPA⁺ game. $\square$

4 OW-FuncCPA

In this section, we formulate a one-way variant of FuncCPA and relate it to the indistinguishability notion when the adversary retains post-challenge access to the functional re-encryption oracle. We define OW-FuncCPA (Definition 7) and show that it implies FuncCPA (Theorem 4), which yields the equivalence stated in Corollary 1. We then show that the weakened variant OW-WeakFuncCPA, where post-challenge access to the functional re-encryption oracle is disallowed, does not imply FuncCPA (Theorem 5).

4.1 Definition of OW-FuncCPA

Definition 7 (OW-FuncCPA Security). Let $\mathcal{E}$ be a public-key encryption scheme with message space $\mathcal{M}$ and a family of functions $\mathcal{F} = \{f : (\mathcal{M} \cup \{\perp\})^\ell \rightarrow \mathcal{M} \mid \ell \in \mathbb{N}\}$. We say that a scheme $\mathcal{E}$ is OW-FuncCPA secure with respect to $\mathcal{F}$ if for all PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ there exists a negligible function $\text{negl}(\cdot)$ such that:

$$\text{Adv}_{\mathcal{A}}^{\text{OW-FuncCPA}}(\lambda) := \Pr[\text{EXP}_{\mathcal{A}, \mathcal{E}, \mathcal{F}}^{\text{OW-FuncCPA}}(\lambda) = 1] = \text{negl}(\lambda)$$

where the OW-FuncCPA experiment $\text{EXP}_{\mathcal{A}, \mathcal{E}, \mathcal{F}}^{\text{OW-FuncCPA}}(\lambda)$ is defined as follows:

1. $\mathcal{CH}$ runs $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$ and gives pk to $\mathcal{A}_1$.
 2. $\mathcal{A}_1$ is given access to a functional re-encryption oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$.
 3. $\mathcal{A}_1$ outputs a challenge request and its state st .
 4. $\mathcal{CH}$ computes the challenge ciphertext $ct^* \leftarrow \text{Enc}(pk, m)$ where $m \xleftarrow{\$} \mathcal{M}$ and gives (ct^*, st) to $\mathcal{A}_2$.
 5. $\mathcal{A}_2$ retains access to the oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$.
 6. $\mathcal{A}_2$ outputs a message $m' \in \mathcal{M}$.
 7. The experiment's output is 1 if $m = m'$, and 0 otherwise.
- **OW-FuncCPA.** When we omit to specify the function family $\mathcal{F}$, it is understood to be the set of all efficiently computable functions.
 - **OW-WeakFuncCPA.** We define the OW-WeakFuncCPA game by modifying the OW-FuncCPA security game: the functional re-encryption oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$ returns $\perp$ for any query made by $\mathcal{A}_2$.

4.2 OW-FuncCPA and FuncCPA are Equivalent

In this section, we show the security notions based on the indistinguishability and the one-wayness are equivalent given access to the functional re-encryption oracle after the challenge phase.

Theorem 4. OW-FuncCPA security implies FuncCPA security.

Proof. Let $\mathcal{E} = (\text{Gen}, \text{Enc}, \text{Dec})$ be a OW-FuncCPA secure encryption scheme defined over a message space $\mathcal{M}$ with message length represented by a polynomial μ . We show that OW-FuncCPA security implies FuncCPA security by constructing a reduction algorithm $\mathcal{B}$ against OW-FuncCPA given an adversary $\mathcal{A}$ for the FuncCPA game.

Reduction algorithm Using a FuncCPA adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, we construct a reduction algorithm $\mathcal{B}$ against OW-FuncCPA as follows:

1. $\mathcal{B}$ receives pk and forwards it to the adversary $\mathcal{A}_1$.
2. For each oracle query (ct_i, f_i) from $\mathcal{A}_1$, $\mathcal{B}$ forwards it to the FuncCPA challenger $\mathcal{CH}$ and relays to $\mathcal{A}_1$ the reply $\text{Enc}(pk, f_i(\text{Dec}(sk, ct_i)))$, using fresh independent randomness in Enc each time.

3. When $\mathcal{A}_1$ outputs messages (m_0, m_1) and a state st , $\mathcal{B}$ submits a challenge request and receives a challenge ciphertext $ct^* = \text{Enc}(pk, m)$ for $m \xleftarrow{\$} \mathcal{M}$.
4. For each $i \in [\mu]$, $\mathcal{B}$ recovers the i -th bit of $f(\text{Dec}(sk, ct))$ as follows. Let n be a repetition parameter that is polynomial in λ ; its value is fixed later in the analysis.
 - (a) For each $j \in [n]$, sample a random mask $\beta_{ij} \leftarrow \{0, 1\}$ and define

$$g_i(x) = \begin{cases} m_0 & \text{if } [x]_i \oplus \beta_{ij} = 0 \\ m_1 & \text{if } [x]_i \oplus \beta_{ij} = 1 \end{cases}.$$

- (b) Using the OW-FuncCPA challenger's oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$, compute

$$ct_{ij}^* \leftarrow \text{reEnc}(pk, sk, ct, g_i \circ \beta_{ij}),$$

where each call uses fresh independent randomness r_{ij} .

- (c) Run the adversary $b'_{ij} \leftarrow \mathcal{A}_2^{\text{reEnc}(pk, sk, \cdot, \cdot)}(ct_{ij}^*, st)$ and set the unmasked bit $b_{ij} \leftarrow b'_{ij} \oplus \beta_{ij}$.
 - (d) Let the guess for the i -th bit be $[m_1]_i \leftarrow \text{maj}(\{b_{ij}\}_{j \in [n]})$.
5. Finally, $\mathcal{B}$ outputs the recovered message $m' = [m']_1 || \dots || [m']_\mu$.

Analysis of the Success Probability Let $\varepsilon := \text{Adv}_{\mathcal{A}}^{\text{FuncCPA}}(\lambda)$ be the advantage of $\mathcal{A}$. We assume that

$$\Pr_{r, b, R} [b = \mathcal{A}_2^{\text{reEnc}(pk, sk, \cdot, \cdot)}(\text{Enc}(pk, m_b; r), st)] = \frac{1}{2} + \frac{\varepsilon}{2}.$$

Here, r denotes the randomness used to encrypt the challenge ciphertext. Let R denote the collection of all remaining randomness used throughout the $\Delta\text{FuncCPA}^+$ game: the randomness used by Gen ; the randomness used by all functional re-encryption oracles $\text{reEnc}(pk, sk, \cdot, \cdot)$ except for the challenge query; the randomness used by $\mathcal{A}_1$; and the randomness used by $\mathcal{A}_2$. Note that st includes the randomness of $\mathcal{A}_2$. We define the subset of $\{R\}$ as follows:

$$\text{Good} = \left\{ R \mid \Pr_{r, b} [m = \mathcal{A}_2^{\text{reEnc}(pk, sk, \cdot, \cdot)}(\text{Enc}(pk, m_b; r), st)] \geq \frac{1}{2} + \frac{\varepsilon}{4} \right\},$$

where $m_0 = 0^\mu, m_1 = f(\text{Dec}(sk, ct^*))$. We show that the subset **Good** is sufficiently large. The proof is the same as in the proof of Lemma 5.

Next, we show if the randomness is good, then recovers $m' = f(\text{Dec}(sk, ct^*))$ correctly in step 4, except negligible probability. The proof is the same as in the proof of Lemma 6.

Now we are ready to bound the success probability of $\mathcal{B}$ as follows:

$$\begin{aligned} & \Pr[\text{EXP}_{\mathcal{B}, \varepsilon}^{\text{OW-FuncCPA}}(\lambda) = 1] \\ &= \Pr[R \in \text{Good}] \Pr[m = m' | R \in \text{Good}] + \Pr[R \notin \text{Good}] \Pr[m = m' | R \notin \text{Good}] \\ &\geq \frac{\varepsilon}{4} \left(1 - \mu \exp \left(-\frac{\varepsilon^2}{8(2 + \varepsilon)} n \right) \right). \end{aligned}$$

Therefore, we have

$$\text{Adv}_{\mathcal{A}}^{\text{FuncCPA}}(\lambda) \leq 2 \cdot \text{Adv}_{\mathcal{B}}^{\text{OW-FuncCPA}}(\lambda) + \text{negl}(\lambda).$$

□

Combining the above theorems, we obtain the following corollary.

Corollary 1. *WeakFuncCPA security, OW-FuncCPA, FuncCPA security and FuncCPA⁺ security are equivalent.*

4.3 OW-WeakFuncCPA Does Not Imply FuncCPA

Theorem 5. *If there exists a OW-CCA2 secure public-key encryption scheme, then OW-WeakFuncCPA security does not imply FuncCPA security.*

Proof. The proof is obtained by combining the following simple facts:

- If there exists an OW-CCA1 secure public-key encryption scheme, we can construct a public-key encryption scheme that is OW-CCA1 secure but not IND-CPA secure.
- OW-CCA1 security implies OW-WeakFuncCPA security. This holds because the functional re-encryption oracle in the OW-WeakFuncCPA game is simulated by the decryption oracle in the OW-CCA1 game.
- FuncCPA security implies IND-CPA security.

Therefore, if OW-WeakFuncCPA security implies FuncCPA security, then OW-CCA1 implies OW-WeakFuncCPA, FuncCPA, and eventually IND-CPA security, which is a contradiction. □

One might mistakenly think that a similar line of reasoning would show that OW-FuncCPA security does not imply FuncCPA security. However, this is not the case since OW-CCA2 security does not imply OW-FuncCPA security. This is because in OW-FuncCPA game, the adversary is allowed to query the challenge ciphertext to the functional re-encryption oracle, unlike OW-CCA2 game.

5 ReEncCPA⁺ Does Not Imply 1-FuncCPA

In this section, we clarify that 1-FuncCPA security is a strictly stronger notion than ReEncCPA security. From this fact, we can obtain two separation results: 1-FuncCPA security is a strictly stronger notion than ReEncCPA security, 1-FuncCPA⁺ security is a strictly stronger notion than ReEncCPA⁺ security.

Theorem 6. *If there exists an IND-CCA2 secure public-key encryption scheme $\mathcal{E}$, then there exists a scheme that is ReEncCPA⁺ secure but not 1-FuncCPA secure.*

Proof. To prove Theorem 6, we construct an encryption scheme $\mathcal{E}'$ which is ReEncCPA^+ secure but not 1-FuncCPA secure. Let $\mathcal{E} = (\text{Gen}, \text{Enc}, \text{Dec})$ be an IND-CCA2 secure public-key encryption scheme over the message space $\{0, 1\}^\lambda$. Note that if there exists an IND-CCA2 secure scheme, there exists a one-way function h .

We construct the scheme $\mathcal{E}'$ over the message space $\{0, 1\}^\lambda$ as follows:

- $\text{Gen}'(1^\lambda) \rightarrow (pk', sk') = ((pk, y, h), (sk, r))$,
 where $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$, $r \xleftarrow{\$} \{0, 1\}^{\lambda-1}$, $h : \{0, 1\}^{\lambda-1} \rightarrow \{0, 1\}^\rho$ is a one-way function (ρ is a polynomial of λ), $y = h(r)$.
- $\text{Enc}'(pk', (t, m)) = \begin{cases} (1, m) & \text{if } y = h(m) \wedge t = 1 \\ (0, \text{Enc}(pk, (t, m))) & \text{otherwise} \end{cases}$,
 where $m \in \{0, 1\}^{\lambda-1}$ and $t \in \{0, 1\}$.
- $\text{Dec}'(sk', (\beta, ct)) = \begin{cases} \text{Dec}(sk, ct) & \text{if } \beta = 0, \\ (1, ct) & \text{if } \beta = 1 \wedge y = h(ct), \\ (0, r) & \text{if } \beta = 1 \wedge y \neq h(ct) \end{cases}$,
 where $\beta \in \{0, 1\}$, $|ct| = \lambda - 1$ if $\beta = 1$, and $|ct| = |\text{Enc}(pk, *)|$ if $\beta = 0$.

As a first step toward the separation, we normalize the adversary's interaction with the functional re-encryption oracle: we can assume, without loss of generality, that the adversary never queries the oracle on ciphertexts previously returned by the oracle. The following lemma shows this assumption.

Lemma 9. *We can assume that a 1-FuncCPA⁺ adversary does not re-query a ciphertext that was a response from the functional re-encryption oracle*

Proof. Let $\mathcal{E}$ be a 1-FuncCPA⁺ secure encryption scheme defined over a message space $\mathcal{M}$. Let $\mathcal{A}$ be an adversary that re-queries responses from the functional re-encryption oracle. We construct an adversary $\mathcal{B}$ that does not re-query such responses.

1. $\mathcal{B}$ receives pk and initializes a list Q to store tuples (x_i, f_i, y_i) of oracle queries made by $\mathcal{A}$ and their responses. Here x_i is a ciphertext, f_i is a function, and $y_i = \text{Enc}(pk, f_i(\text{Dec}(sk, x_i)))$.
2. For each oracle query (x_i, f_i) from $\mathcal{A}$, $\mathcal{B}$ returns as follows:
 - (a) If $x_i \notin \{y_j \mid (x_j, f_j, y_j) \in Q, j < i\}$, $\mathcal{B}$ forwards (x_i, f_i) to the 1-FuncCPA⁺ challenger $\mathcal{CH}$, and receives the response y_i . $\mathcal{B}$ adds (x_i, f_i, y_i) to Q and returns y_i to $\mathcal{A}$.
 - (b) If $x_i \in \{y_j \mid (x_j, f_j, y_j) \in Q, j < i\}$, let $j < i$ be such that $(x_j, f_j, y_j) \in Q$ and $y_j = x_i$. Then $\mathcal{B}$ queries its challenger on $(x_j, f_i \circ f_j)$, receives the response y_i , appends (x_i, f_i, y_i) to Q , and returns y_i to $\mathcal{A}$.
3. $\mathcal{B}$ outputs $b' \leftarrow \mathcal{A}$

We check that $\mathcal{B}$ never re-queries a ciphertext previously returned by the oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$. Consider any oracle query (x_i, f_i) issued by $\mathcal{A}$. If $x_i \notin \{y_j \mid (x_j, f_j, y_j) \in Q, j < i\}$, that is, x_i is not equal to any prior oracle response, then $\mathcal{B}$ forwards (x_i, f_i) to $\mathcal{CH}$. Otherwise, let $j < i$ with $(x_j, f_j, y_j) \in Q$ and

$y_j = x_i$; then $\mathcal{B}$ queries $\mathcal{CH}$ on $(x_j, f_i \circ f_j)$ instead of (x_i, f_i) , composing f_j into the current request and avoiding a re-query. By correctness, $\text{Dec}(sk, y_j) = f_j(\text{Dec}(sk, x_j))$, and hence

$$f_i(\text{Dec}(sk, y_j)) = f_i(f_j(\text{Dec}(sk, x_j))) = (f_i \circ f_j)(\text{Dec}(sk, x_j)).$$

Therefore, the response to $(x_j, f_i \circ f_j)$ encrypts the same plaintext as the re-query (x_i, f_i) would have produced, while x_j is not an oracle output to $\mathcal{B}$. In all cases, $\mathcal{B}$ avoids submitting to $\mathcal{CH}$ any ciphertext that originated from $\mathcal{CH}$, and the simulated responses to $\mathcal{A}$ are correct. $\square$

Lemma 10. $\mathcal{E}'$ is ReEncCPA^+ secure.

Proof. We proceed by a standard hybrid argument. Let Q be the number of oracle queries made by $\mathcal{A}$ in the ReEncCPA^+ experiment. For each $i \in \{0, \dots, Q\}$, define the hybrid experiment H_i as follows.

H_i . In this hybrid, the challenger answers the first i oracle queries by returning $\text{Enc}'(pk', 0)$, and answers the remaining queries using the functional re-encryption oracle $\text{reEnc}(pk, sk, \cdot, \cdot)$. We now focus on the response to the i -th oracle query. For the i -th oracle query (β_i, ct_i) , the challenger responds as follows:

- If $\beta_i = 0$, compute $\text{Dec}(sk, ct_i) = (t, m)$ with $t \in \{0, 1\}$ and $|m| = \lambda - 1$; returns $(1, m)$ if $t = 1$ and $y = h(m)$, and $(0, \text{Enc}(pk, (t, m)))$ otherwise.
- If $\beta_i = 1$, returns $(1, ct_i)$ if $y = h(ct_i)$, and $(0, \text{Enc}(pk, (r, 0)))$ otherwise.

Equivalently, the response equals $\text{Enc}'(pk', \text{Dec}'(sk', (\beta_i, ct_i)))$. By construction, H_0 coincides with $\text{EXP}_{\mathcal{A}, \mathcal{E}'}^{\text{ReEncCPA}^+}(\lambda)$ when $b = 1$, and H_Q coincides with $\text{EXP}_{\mathcal{A}, \mathcal{E}'}^{\text{ReEncCPA}^+}(\lambda)$ when $b = 0$.

We show that H_i and H_{i-1} are indistinguishable via the following modifications to the response to the i -th query:

1. The response is changed from $\text{Enc}'(pk', m_i)$ to $(0, \text{Enc}(pk, m_i))$, where $(t_i, m_i) \leftarrow \text{Dec}'(sk', (\beta_i, ct_i))$; this change follows from the one-wayness of h , which implies that the event $\text{Dec}'(sk', (\beta_i, ct_i)) = (1, r)$ is negligible.
2. The response to the i -th query is replaced by the fixed pair $(0, \text{Enc}(pk, 0))$; distinguishing these hybrids reduces to breaking the IND-CCA2 security of $\mathcal{E}$.

To show that the probability of the i -th queried ciphertext decrypting to r is negligible, we introduce a new game H'_i between games H_i and H_{i-1} .

H'_i . This hybrid differs only in the response to the i -th query. For the i -th query (β_i, ct_i) , the challenger answers:

- If $\beta_i = 0$, compute $\text{Dec}(sk, ct_i) = (t, m)$ with $t \in \{0, 1\}$ and $|m| = \lambda - 1$, and return $(0, \text{Enc}(pk, (t, m)))$
- If $\beta_i = 1$, return $(0, \text{Enc}(pk, (r, 0)))$.

Lemma 11.

$$|\Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\text{H}_i}(\lambda) = 1] - \Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\text{H}'_i}(\lambda) = 1]| = \text{negl}(\lambda)$$

Proof. For the i -th query in Game H_i , we define the bad event Bad_i as follows.

Bad _{i} Let $(t_i, m_i) := \text{Dec}'(sk', (\beta_i, ct_i))$ with $t_i \in \{0, 1\}$ and $m_i \in \{0, 1\}^{\lambda-1}$. The event occurs when $t_i = 1$ and $y = h(m_i)$.

By the definition of Bad_i , the two hybrids coincide on the event $\neg \text{Bad}_i$. Hence

$$\Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\text{H}_i}(\lambda) = 1 \wedge \neg \text{Bad}_i] = \Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\text{H}'_i}(\lambda) = 1 \wedge \neg \text{Bad}_i].$$

Therefore,

$$|\Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\text{H}_i}(\lambda) = 1] - \Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\text{H}'_i}(\lambda) = 1]| \leq \Pr[\text{Bad}_i].$$

It remains to show that $\Pr[\text{Bad}_i]$ is negligible. To prove this, we construct a reduction $\mathcal{B}_i$ that uses $\mathcal{A}$ to invert h .

1. $\mathcal{B}_i$ receives $y = h(r)$ from the OWF challenger, samples $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$, sets $pk' \leftarrow (pk, y)$, and gives pk' to $\mathcal{A}$. Note that $\mathcal{B}_i$ does not know r .
2. $\mathcal{B}_i$ answers oracle queries (β_j, ct_j) from $\mathcal{A}$ as follows:
 - (a) If $j < i$, return $(0, \text{Enc}(pk, 0))$.
 - (b) If $j = i$, proceed and output to the OWF challenger:
 - i. If $\beta_i = 0$, compute $\text{Dec}(sk, ct_i) = (t, m)$ with $t \in \{0, 1\}$ and $|m| = \lambda - 1$; output m .
 - ii. If $\beta_i = 1$, output ct_i .

We bound the probability that the bad event Bad_i occurs by the reduction's success probability, namely the probability of breaking the one-wayness of h . For the first $i - 1$ queries (β_j, ct_j) , $\mathcal{B}_i$ answers exactly as in H_i , so the simulation is perfect up to the $(i - 1)$ -th query. Consider the i -th query (β_i, ct_i) .

- If $\beta_i = 0$, compute $\text{Dec}(sk, ct_i) = (t, m)$ with $t \in \{0, 1\}$ and $|m| = \lambda - 1$; Bad_i occurs iff $t = 1$ and $y = h(m)$. In this case, $\mathcal{B}_i$ outputs m , which is a valid preimage of y .
- If $\beta_i = 1$, then, by the definition of Dec' , Bad_i occurs iff $y = h(ct_i)$. In that case, $\mathcal{B}_i$ outputs ct_i , which is a valid preimage of y .

Therefore, if Bad_i occurs, $\mathcal{B}_i$ outputs a valid preimage of y , i.e., $\mathcal{B}_i$ breaks the one-wayness of h . Hence, $\Pr[\text{Bad}_i]$ is upper-bounded by the advantage of $\mathcal{B}_i$ in breaking the one-wayness of h , which is negligible in λ . That is, $\Pr[\text{Bad}_i] = \text{negl}(\lambda)$. $\square$

Next, we change the response of the i -th query to $(0, \text{Enc}(pk, 0))$ by distinguishing between H'_i and H_{i-1} , which reduces to breaking the IND-CCA2 security of $\mathcal{E}$.

Lemma 12.

$$|\Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\text{H}'_i}(\lambda) = 1] - \Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\text{H}_{i-1}}(\lambda) = 1]| = \text{negl}(\lambda)$$

Proof. We build an IND-CCA2 adversary $\mathcal{B}'_i$ against $\mathcal{E}$ that uses $\mathcal{A}$ as a subroutine. By Lemma 9, $\mathcal{A}$ never re-queries oracle outputs, so $\mathcal{B}'_i$ never asks its decryption oracle on the challenge ciphertext.

1. $\mathcal{B}'_i$ receives pk from the IND-CCA2 challenger $\mathcal{CH}$. $\mathcal{B}'_i$ samples $r \xleftarrow{\$} \{0,1\}^{\lambda-1}$, sets $y \leftarrow h(r)$, and gives $pk' = (pk, y)$ to $\mathcal{A}$.
2. For each oracle query (β_j, ct_j) with $j < i$, $\mathcal{B}'_i$ returns $(0, \text{Enc}(pk, 0))$.
3. For the i -th query (β_i, ct_i) :
 - (a) If $\beta_i = 0$, $\mathcal{B}'_i$ queries the decryption oracle on ct_i to obtain $(t, m) = \text{Dec}(sk, ct_i)$ and sets $m_1 \leftarrow (t, m)$.
 - (b) If $\beta_i = 1$, it sets $m_1 \leftarrow (0, r)$.
Let $m_0 = 0^\lambda$. $\mathcal{B}'_i$ submits (m_0, m_1) to $\mathcal{CH}$, receives the challenge ciphertext $ct^* = \text{Enc}(pk, m_b)$, and returns $(0, ct^*)$ to $\mathcal{A}$ as the response to the i -th query.
4. For each further query (β_j, ct_j) with $j > i$:
 - (a) If $\beta_j = 0$, $\mathcal{B}'_i$ queries the decryption oracle on ct_j to get $(t_j, m_j) = \text{Dec}(sk, ct_j)$.
 - (b) If $\beta_j = 1$ and $y = h(ct_j)$, it sets $(t_j, m_j) \leftarrow (1, ct_j)$; otherwise it sets $(t_j, m_j) \leftarrow (0, r)$.
Then it returns $\text{Enc}'((pk, y), (t_j, m_j))$.
5. Finally, when $\mathcal{A}$ outputs a bit b' , $\mathcal{B}'_i$ outputs the same b' .

We now argue that the view of $\mathcal{A}$ in our reduction is identically distributed to its view in the target hybrids. When the IND-CCA2 challenger bit is $b = 1$, the i -th response returned by $\mathcal{B}'_i$ is $(0, \text{Enc}(pk, m_1))$, which coincides with the rule of $\mathcal{H}'_i$. When $b = 0$, the i -th response is $(0, \text{Enc}(pk, 0))$, matching $\mathcal{H}_{i-1}$. For all $j \neq i$, $\mathcal{B}'_i$ answers exactly as in the corresponding hybrid by construction: the first $i - 1$ queries are returned as $(0, \text{Enc}(pk, 0))$, and the remaining queries are answered according to Enc' , using the decryption oracle only where permitted. By Lemma 9, the adversary never re-submits the i -th response, so the challenge ciphertext ct^* is never queried to the decryption oracle. Hence any distinguisher between $\mathcal{H}'_i$ and $\mathcal{H}_{i-1}$ yields an IND-CCA2 adversary against $\mathcal{E}$ with the same advantage; formally,

$$|\Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\mathcal{H}'_i}(\lambda) = 1] - \Pr[\text{EXP}_{\mathcal{E}', \mathcal{A}}^{\mathcal{H}_{i-1}}(\lambda) = 1]| \leq \text{Adv}_{\mathcal{E}, \mathcal{B}'_i}^{\text{IND-CCA2}}(\lambda).$$

□

From the Lemmas 11 and 12, it follows that $\mathcal{E}'$ is ReEncCPA^+ secure. □

Lemma 13. $\mathcal{E}'$ is not 1-FuncCPA secure.

Proof. We construct a 1-FuncCPA adversary $\mathcal{A}$ against $\mathcal{E}'$. The 1-FuncCPA computes $(pk', sk') \leftarrow \text{Gen}'(\lambda)$, samples $b \xleftarrow{\$} \{0,1\}$ and gives pk' to $\mathcal{A}$.

1. $\mathcal{A}$ receives pk' from the $\mathcal{CH}$.
2. $\mathcal{A}$ defines $f(x) = (1, x')$ where $x = (t, x')$ with $t \in \{0,1\}$ and $|x'| = \lambda - 1$, and queries (ct, f) with $ct = (1, 1^{\lambda-1})$ to the $\text{reEnc}(pk', sk', \cdot, \cdot)$ oracle.

3. $\mathcal{CH}$ computes $\text{Dec}'(sk', ct)$. By the definition of Dec' , $\text{Dec}'(sk', (1, 1^{\lambda-1})) = (0, r)$. Therefore, $f((0, r)) = (1, r)$, and $ct' = \text{Enc}'(pk', (1, r)) = (1, r)$ is returned.
4. $\mathcal{A}$ submits the challenge query $m_0 = (0, 0^{\lambda-1})$ and $m_1 = (1, r)$, and then receives the challenge ciphertext $(\beta^*, ct^*) = \text{Enc}'(pk', m_b)$.
5. $\mathcal{A}$ outputs β^* .

We show that the adversary $\mathcal{A}$ predicts the challenger's bit b with non-negligible probability. If $b = 0$, the challenger responds to the challenge query $(m_0 = (0, 0^{\lambda-1}), m_1 = (1, r))$ with $(\beta^*, ct^*) = (0, \text{Enc}(pk, (0, 0^{\lambda-1})))$. The adversary $\mathcal{A}$ then outputs β^* , hence outputs 0. Similarly, if $b = 1$, the challenger returns $(\beta^*, ct^*) = (1, r)$, and $\mathcal{A}$ outputs 1. Therefore $\mathcal{A}$ correctly guesses the challenger's bit b with non-negligible probability, which shows that $\mathcal{E}'$ is not 1-FuncCPA-secure. $\square$

From the above Lemmas 10 and 13, we obtain the Theorem 6. $\square$

It follows from Theorem 6 and the fact that 1-FuncCPA security implies ReEncCPA security that we obtain Corollary 2.

Corollary 2. *If an IND-CCA2-secure public-key encryption scheme $\mathcal{E}$ exists, then there exists a scheme that is ReEncCPA secure but not 1-FuncCPA secure.*

Similarly, it follows from Theorem 6 and the fact that 1-FuncCPA⁺ security implies ReEncCPA⁺ security that we obtain Corollary 3.

Corollary 3. *If an IND-CCA2-secure public-key encryption scheme $\mathcal{E}$ exists, then there exists a scheme that is ReEncCPA⁺ secure but not 1-FuncCPA⁺ secure.*

References

1. Akavia, A., Gentry, C., Halevi, S., Vald, M.: Achievable CCA2 relaxation for homomorphic encryption. In: Kiltz, E., Vaikuntanathan, V. (eds.) TCC 2022, Part II. LNCS, vol. 13748, pp. 70–99. Springer, Cham (Nov 2022). https://doi.org/10.1007/978-3-031-22365-5_3
2. Akavia, A., Gentry, C., Halevi, S., Vald, M.: Achievable CCA2 relaxation for homomorphic encryption. J. Cryptol. **38**(1), 5 (2025). <https://doi.org/10.1007/S00145-024-09526-1>, <https://doi.org/10.1007/s00145-024-09526-1>
3. Brakerski, Z., Goldreich, O.: From Absolute Distinguishability to Positive Distinguishability, pp. 141–155. Springer Berlin Heidelberg, Berlin, Heidelberg (2011). https://doi.org/10.1007/978-3-642-22670-0_17
4. Dodis, Y., Halevi, S., Wichs, D.: Security with functional re-encryption from CPA. In: Rothblum, G.N., Wee, H. (eds.) TCC 2023, Part II. LNCS, vol. 14370, pp. 279–305. Springer, Cham (Nov / Dec 2023). https://doi.org/10.1007/978-3-031-48618-0_10
5. Dodis, Y., Yu, Y.: Overcoming weak expectations. In: Sahai, A. (ed.) TCC 2013. LNCS, vol. 7785, pp. 1–22. Springer, Berlin, Heidelberg (Mar 2013). https://doi.org/10.1007/978-3-642-36594-2_1

6. Manulis, M., Nguyen, J.: Fully homomorphic encryption beyond IND-CCA1 security: Integrity through verifiability. In: Joye, M., Leander, G. (eds.) *Advances in Cryptology - EUROCRYPT 2024 - 43rd Annual International Conference on the Theory and Applications of Cryptographic Techniques*, Zurich, Switzerland, May 26-30, 2024, Proceedings, Part II. *Lecture Notes in Computer Science*, vol. 14652, pp. 63–93. Springer (2024). https://doi.org/10.1007/978-3-031-58723-8_3, https://doi.org/10.1007/978-3-031-58723-8_3
7. Nakajima, E., Hara, K., Yamashita, K.: On multi-key func CPA secure encryption schemes. In: *2024 1st International Conference On Cryptography And Information Security (VCRIS)*. pp. 1–6 (2024). <https://doi.org/10.1109/VCRIS63677.2024.10813389>
8. Nuida, K.: How to handle invalid queries for malicious-private protocols based on homomorphic encryption. In: *Proceedings of the 9th ACM on ASIA Public-Key Cryptography Workshop*. pp. 15–25 (2022)